ACTIVITY SHEET – 6

Programs Using Method Overloading and Method Overriding

1. OBJECTIVES

By completing this activity, students will be able to:

• Understand compile-time polymorphism using method overloading
• Understand runtime polymorphism using method overriding
• Apply polymorphism to solve logical and industry-oriented problems
• Design reusable object-oriented programs
• Analyze scenarios and identify when overloading or overriding is appropriate

2. READING REQUIREMENTS

Oracle Java Tutorials – Method Overloading:
https://docs.oracle.com/javase/tutorial/java/javaOO/methods.html

Oracle Java Tutorials – Method Overriding:
https://docs.oracle.com/javase/tutorial/java/IandI/override.html

3. ILLUSTRATIVE EXAMPLES

Example 1: Method Overloading

Code:

```
class Calculator {
   int add(int a, int b){
      return a + b;
   }

   int add(int a, int b, int c){
      return a + b + c;
   }

   double add(double a, double b){
      return a + b;
   }
}
```

Java Code:

Welcome | CalculatorTest.java

CalculatorTest.java > CalculatorTest > main(String[])

```java
import java.util.Scanner;

class Calculator {
    int add(int a, int b){
        return a + b;
    }

    int add(int a, int b, int c){
        return a + b + c;
    }

    double add(double a, double b){
        return a + b;
    }
}

public class CalculatorTest {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        Calculator calc = new Calculator();

        System.out.println(x: "Enter two integers:");
        int a = sc.nextInt();
        int b = sc.nextInt();
        System.out.println("Sum of two integers: " + calc.add(a, b));
```

Welcome | CalculatorTest.java

CalculatorTest.java > CalculatorTest

```java
public class CalculatorTest {
    public static void main(String[] args) {

        System.out.println(x: "Enter three integers:");
        int x = sc.nextInt();
        int y = sc.nextInt();
        int z = sc.nextInt();
        System.out.println("Sum of three integers: " + calc.add(x, y, z));

        System.out.println(x: "Enter two decimal numbers:");
        double d1 = sc.nextDouble();
        double d2 = sc.nextDouble();
        System.out.println("Sum of two doubles: " + calc.add(d1, d2));

        sc.close();
    }
}
```

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\sheet.java>javac CalculatorTest.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\sheet.java>java CalculatorTest.java
Enter two integers:
1
2
Sum of two integers: 3
Enter three integers:
1
2
3
Sum of three integers: 6
Enter two decimal numbers:
2.3
5.5
Sum of two doubles: 7.8
```

Example 2: Method Overriding

Code:

```
class Animal{
 void sound(){
 System.out.println("Animals make sound");
 }
}

class Dog extends Animal{
 void sound(){
 System.out.println("Dog barks");
 }
}
```

Java Code:

Welcome | CalculatorTest.java | DogTest.java

DogTest.java > ...

```
class Animal{
    void sound(){
        System.out.println(x: "Animals make sound");
    }
}

class Dog extends Animal{
    void sound(){
        System.out.println(x: "Dog barks");
    }
}

public class DogTest {
    Run | Debug
    public static void main(String[] args) {

        Animal a = new Animal();
        a.sound();

        Dog d = new Dog();
        d.sound();

        Animal ref = new Dog();
        ref.sound();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\sheet.java>javac DogTest.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\sheet.java>java DogTest.java
Animals make sound
Dog barks
Dog barks
```

1. A printer application needs to display different types of information. Sometimes it prints a number, sometimes a text message, and sometimes both number and text. Design a Java program so the printer performs all these tasks using the same method name.
Hint: Behavior should depend on different parameters.

JAVA CODE:

Main_Printer.java | NotificationSystem.java | LoggingSystem.java

```java
import java.util.Scanner;
class Printer {
    void print(int number) {
        System.out.println("Number: " + number);
    }
    void print(String message) {
        System.out.println("Message: " + message);
    }
    void print(int number, String message) {
        System.out.println("Number: " + number + ", Message: " + message);
    }
}
public class Main_Printer {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Printer p = new Printer();
        System.out.println(x: "Choose an option:");
        System.out.println(x: "1. Print number");
        System.out.println(x: "2. Print text");
        System.out.println(x: "3. Print number and text");
        System.out.print(s: "Enter your choice: ");
        int choice = sc.nextInt();
        sc.nextLine();
```

Main_Printer.java | NotificationSystem.java | LoggingSystem.java

```java
public class Main_Printer {
    public static void main(String[] args) {
        if (choice == 1) {
            System.out.print(s: "Enter a number: ");
            int num = sc.nextInt();
            p.print(num);
        }
        else if (choice == 2) {
            System.out.print(s: "Enter a text message: ");
            String msg = sc.nextLine();
            p.print(msg);
        }
        else if (choice == 3) {
            System.out.print(s: "Enter a number: ");
            int num = sc.nextInt();
            sc.nextLine();
            System.out.print(s: "Enter a text message: ");
            String msg = sc.nextLine();
            p.print(num, msg);
        }
        else {
            System.out.println(x: "Invalid choice!");
        }
        sc.close();
    }
}
```

OUTPUT:

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Printer.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Printer.java
Choose an option:
1. Print number
2. Print text
3. Print number and text
Enter your choice: 1
Enter a number: 16
Number: 16

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Printer.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Printer.java
Choose an option:
1. Print number
2. Print text
3. Print number and text
Enter your choice: 2
Enter a text message: PAVANI REDDY
Message: PAVANI REDDY
```

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Printer.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Printer.java
Choose an option:
1. Print number
2. Print text
3. Print number and text
Enter your choice: 3
Enter a number: 16
Enter a text message: PAVANI
Number: 16, Message: PAVANI

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Printer.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Printer.java
Choose an option:
1. Print number
2. Print text
3. Print number and text
Enter your choice: 4
Invalid choice!
```

2. A notification system sends alerts to users. A base class Notification sends a general notification, while EmailNotification sends email alerts. Design a program where the notification behavior changes when called from the child class.
Hint: Child class should redefine the method behavior.

JAVA CODE:

Main_Printer.java | Main_Notification_System.java | LoggingSystem.java

Main_Printer.java | Main_Notification_System.java | LoggingSystem.java

Main_Notification_System.java > ...

```java
public class Main_Notification_System {
    public static void main(String[] args) {
        System.out.print(s: "Enter notification message: ");
        String message = sc.nextLine();

        Notification n;

        if (choice == 1) {
            n = new Notification();
            n.sendNotification(message);
        }
        else if (choice == 2) {
            System.out.print(s: "Enter email address: ");
            String email = sc.nextLine();

            EmailNotification en = new EmailNotification();
            en.sendEmailNotification(email, message);
        }
        else {
            System.out.println(x: "Invalid choice!");
        }

        sc.close();
    }
}

        System.out.println(x: "Choose notification type:");
        System.out.println(x: "1. General Notification");
        System.out.println(x: "2. Email Notification");
        System.out.print(s: "Enter your choice: ");
        int choice = sc.nextInt();
        sc.nextLine();
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Notification_System.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Notification_System.java
Choose notification type:
1. General Notification
2. Email Notification
Enter your choice: 1
Enter notification message: happy birthday
General Notification Sent: happy birthday

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Notification_System.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Notification_System.java
Choose notification type:
1. General Notification
2. Email Notification
Enter your choice: 2
Enter notification message: hi
Enter email address: pavani2007@gmail.com
Email sent to pavani2007@gmail.com: hi
```

3. A logging system records system events. Sometimes it logs only a message, sometimes message with timestamp, and sometimes message with timestamp and error code. Design a program where the same method name logs all these cases.
Hint: Same method name with different parameter lists.

JAVA CODE:

Main_Printer.java | Main_Notification_System.java | Main_Logger.java

Main_Logger.java > Main_Logger

```java
import java.util.Scanner;
class Logger {
    void log(String message) {
        System.out.println("Log Message: " + message);
    }
    void log(String message, String timestamp) {
        System.out.println("Log Message: " + message + " | Timestamp: " + timestamp);
    }
    void log(String message, String timestamp, int errorCode) {
        System.out.println("Log Message: " + message + " | Timestamp: " + timestamp + " | Error Code: " + errorCode);
    }
}
public class Main_Logger {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Logger l = new Logger();

        System.out.println(x: "Choose logging option:");
        System.out.println(x: "1. Log only message");
        System.out.println(x: "2. Log message with timestamp");
        System.out.println(x: "3. Log message with timestamp and error code");
        System.out.print(s: "Enter your choice: ");
        int choice = sc.nextInt();
        sc.nextLine();

        if (choice == 1) {
            System.out.print(s: "Enter message: ");
            String message = sc.nextLine();
            l.log(message);
        }
```

Main_Printer.java | Main_Notification_System.java | Main_Logger.java

Main_Logger.java > Main_Logger

```java
public class Main_Logger {
    public static void main(String[] args) {
        else if (choice == 2) {
            System.out.print(s: "Enter message: ");
            String message = sc.nextLine();

            System.out.print(s: "Enter timestamp: ");
            String timestamp = sc.nextLine();

            l.log(message, timestamp);
        }
        else if (choice == 3) {
            System.out.print(s: "Enter message: ");
            String message = sc.nextLine();

            System.out.print(s: "Enter timestamp: ");
            String timestamp = sc.nextLine();

            System.out.print(s: "Enter error code: ");
            int errorCode = sc.nextInt();

            l.log(message, timestamp, errorCode);
        }
        else {
            System.out.println(x: "Invalid choice!");
        }

        sc.close();
    }
}
```

OUTPUT:

```
C:\Windows\System32\cmd.e

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Logger.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Logger.java
Choose logging option:
1. Log only message
2. Log message with timestamp
3. Log message with timestamp and error code
Enter your choice: 1
Enter message: hi
Log Message: hi

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Logger.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Logger.java
Choose logging option:
1. Log only message
2. Log message with timestamp
3. Log message with timestamp and error code
Enter your choice: 2
Enter message: hi
Enter timestamp: 2
Log Message: hi | Timestamp: 2
```

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Logger.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Logger.java
Choose logging option:
1. Log only message
2. Log message with timestamp
3. Log message with timestamp and error code
Enter your choice: 3
Enter message: 3
Enter timestamp: 2
Enter error code: 404
Log Message: 3 | Timestamp: 2 | Error Code: 404

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Logger.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Logger.java
Choose logging option:
1. Log only message
2. Log message with timestamp
3. Log message with timestamp and error code
Enter your choice: 4
Invalid choice!
```

4. Given a large integer represented as an integer array digits where each element represents a digit, add one to the integer and return the updated array. (For Practice)

Example: Input: [2,9,9] Output: [3,0,0]

JAVA CODE:

tem_Process.java ● | Main_Smart_Home.java | Main_Discount_Calculator.java

AddOneToArray.java > ...

```java
import java.util.Scanner;
public class AddOneToArray {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter number of digits: ");
        int n = sc.nextInt();
        int[] digits = new int[n];
        System.out.println(x: "Enter the digits:");
        for (int i = 0; i < n; i++) {
            digits[i] = sc.nextInt();
        }
        int[] result = plusOne(digits);
        System.out.print(s: "Output: [");
        for (int i = 0; i < result.length; i++) {
            System.out.print(result[i]);
            if (i < result.length - 1) {
                System.out.print(s: ",");
            }
        }
        System.out.println(x: "]");
        sc.close();
    }
    public static int[] plusOne(int[] digits) {
        for (int i = digits.length - 1; i >= 0; i--) {
            if (digits[i] < 9) {
                digits[i]++;
                return digits;
            }
            digits[i] = 0;
        }
        int[] newDigits = new int[digits.length + 1];
        newDigits[0] = 1;
        return newDigits;
    }
}
```

OUTPUT:

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac AddOneToArray.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java AddOneToArray.java
Enter number of digits: 5
Enter the digits:
1 2 3 4 5
Output: [1,2,3,4,6]
```

5. Write a Java program to check whether brackets in a string are balanced. The brackets include (), {}, and [].(For Practice)

Example: Input: {[()]} Output: Valid

JAVA CODE:

Main_String_Valid_Parentheses.java > ...

```java
import java.util.*;
public class Main_String_Valid_Parentheses {
    static boolean isBalanced(String s) {
        Stack<Character> st = new Stack<>();
        for (int i = 0; i < s.length(); i++) {
            char ch = s.charAt(i);
            if (ch == '(' || ch == '{' || ch == '[') {
                st.push(ch);
            }
            else if (ch == ')' || ch == '}' || ch == ']') {
                if (st.isEmpty()) return false;
                char top = st.pop();
                if ((ch == ')' && top != '(') ||
                    (ch == '}' && top != '{') ||
                    (ch == ']' && top != '[')) {
                    return false;
                }
            }
        }
        return st.isEmpty();
    }
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter a string with brackets: ");
        String s = sc.nextLine();
        if (isBalanced(s)) {
            System.out.println(x: "Valid");
        } else {
            System.out.println(x: "Invalid");
        }
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_String_Valid_Parentheses.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_String_Valid_Parentheses.java
Enter a string with brackets: {[()]}
Valid
```

6. A payment system processes payments differently depending on whether the payment is made using cash, card, or UPI. Design a class PaymentProcessor where the same method processPayment() handles different payment inputs.
Hint: Method overloading is applicable.

JAVA CODE:

Main_Printer.java | Main_Notification_System.java | Main_Logger.java | Main_String_Valid_P

Main_Payment_System_Process.java > Main_Payment_System_Process > main(String[])

```java
import java.util.Scanner;
class PaymentProcessor {
    void processPayment(double amount) {
        System.out.println("\n--- CASH PAYMENT ---");
        System.out.printf("Amount: ₹%.2f%n", amount);
        System.out.println("Payment Successful (Cash).");
    }
    void processPayment(double amount, String cardNumber, String cardHolder) {
        System.out.println("\n--- CARD PAYMENT ---");
        System.out.printf("Amount: ₹%.2f%n", amount);
        System.out.println("Card Holder: " + cardHolder);
        System.out.println("Card Number (masked): " + maskCard(cardNumber));
        System.out.println("Payment Successful (Card).");
    }
    void processPayment(double amount, String upiId) {
        System.out.println("\n--- UPI PAYMENT ---");
        System.out.printf("Amount: ₹%.2f%n", amount);
        System.out.println("UPI ID: " + upiId);
        System.out.println("Payment Successful (UPI).");
    }
    private String maskCard(String cardNumber) {
        if (cardNumber == null) return "Invalid";
        cardNumber = cardNumber.replaceAll("\\s+", "");
        if (cardNumber.length() < 4) return "XXXX";
        String last4 = cardNumber.substring(cardNumber.length() - 4);
        return "XXXX-XXXX-XXXX-" + last4;
    }
}
public class Main_Payment_System_Process {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        PaymentProcessor pp = new PaymentProcessor();
        System.out.println("Choose Payment Method:");
        System.out.println("1. Cash");
        System.out.println("2. Card");
        System.out.println("3. UPI");
```

Main_Printer.java | Main_Notification_System.java | Main_Logger.java | Main_String_Val

Main_Payment_System_Process.java > ...

```java
public class Main_Payment_System_Process {
    public static void main(String[] args) {
        System.out.print(s: "Enter choice (1/2/3): ");
        int choice = sc.nextInt();
        sc.nextLine();
        System.out.print(s: "Enter amount: ");
        double amount = sc.nextDouble();
        sc.nextLine();
        switch (choice) {
            case 1:
                pp.processPayment(amount);
                break;
            case 2:
                System.out.print(s: "Enter card number: ");
                String cardNumber = sc.nextLine();
                System.out.print(s: "Enter card holder name: ");
                String cardHolder = sc.nextLine();
                pp.processPayment(amount, cardNumber, cardHolder);
                break;
            case 3:
                System.out.print(s: "Enter UPI ID (example: name@bank): ");
                String upiId = sc.nextLine();
                pp.processPayment(amount, upiId);
                break;
            default:
                System.out.println(x: "Invalid choice!");
        }
        sc.close();
    }
}
```

OUTPUT:

```
C:\Windows\System32\cmd.e

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Payment_System_Process.java
Choose Payment Method:
1. Cash
2. Card
3. UPI
Enter choice (1/2/3): 1
Enter amount: 100

--- CASH PAYMENT ---
Amount: ?100.00
Payment Successful (Cash).

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Payment_System_Process.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Payment_System_Process.java
Choose Payment Method:
1. Cash
2. Card
3. UPI
Enter choice (1/2/3): 2
Enter amount: 200
Enter card number: 1234567891234567
Enter card holder name: pavani

--- CARD PAYMENT ---
Amount: ?200.00
Card Holder: pavani
Card Number (masked): XXXX-XXXX-XXXX-4567
Payment Successful (Card).

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Payment_System_Process.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Payment_System_Process.java
Choose Payment Method:
1. Cash
2. Card
3. UPI
Enter choice (1/2/3): 3
Enter amount: 1234
Enter UPI ID (example: name@bank): pavani@SBI

--- UPI PAYMENT ---
Amount: ?1234.00
UPI ID: pavani@SBI
Payment Successful (UPI).
```

7. A smart home application controls devices such as Light, Fan, and AirConditioner. All devices inherit from SmartDevice and implement their own version of turnOn(). Design a Java program to simulate this system.
Hint:Method overriding is applicable.

JAVA CODE:

Main_Notification_System.java ● | Main_Logger.java | Main_String_Valid_Parentheses.java | M

Main_Smart_Home.java > Main_Smart_Home

```java
import java.util.Scanner;
class SmartDevice {
    String deviceName;
    SmartDevice(String deviceName) {
        this.deviceName = deviceName;
    }
    void turnOn() {
        System.out.println(deviceName + " is turned ON.");
    }
}
class Light extends SmartDevice {
    Light() {
        super(deviceName: "Light");
    }
    @Override
    void turnOn() {
        System.out.println(x: "Light is turned ON  (Brightening the room).");
    }
}
class Fan extends SmartDevice {
    Fan() {
        super(deviceName: "Fan");
    }
    @Override
    void turnOn() {
        System.out.println(x: "Fan is turned ON  (Spinning fast).");
    }
}
class AirConditioner extends SmartDevice {
    AirConditioner() {
        super(deviceName: "Air Conditioner");
    }
    @Override
    void turnOn() {
        System.out.println(x: "Air Conditioner is turned ON  (Cooling started).");
    }
}
```

Main_Notification_System.java ● | Main_Logger.java | Main_String_Valid_Parentheses

Main_Smart_Home.java > Main_Smart_Home

```java
public class Main_Smart_Home {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "SMART HOME DEVICE CONTROL");
        System.out.println(x: "1. Light");
        System.out.println(x: "2. Fan");
        System.out.println(x: "3. Air Conditioner");
        System.out.print(s: "Choose a device to turn ON (1/2/3): ");
        int choice = sc.nextInt();
        SmartDevice device;
        switch (choice) {
            case 1:
                device = new Light();
                break;
            case 2:
                device = new Fan();
                break;
            case 3:
                device = new AirConditioner();
                break;
            default:
                System.out.println(x: "Invalid choice!");
                sc.close();
                return;
        }
        device.turnOn();
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Smart_Home.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Smart_Home.java
SMART HOME DEVICE CONTROL
1. Light
2. Fan
3. Air Conditioner
Choose a device to turn ON (1/2/3): 1
Light is turned ON  (Brightening the room).

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Smart_Home.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Smart_Home.java
SMART HOME DEVICE CONTROL
1. Light
2. Fan
3. Air Conditioner
Choose a device to turn ON (1/2/3): 2
Fan is turned ON  (Spinning fast).
```

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Smart_Home.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Smart_Home.java
SMART HOME DEVICE CONTROL
1. Light
2. Fan
3. Air Conditioner
Choose a device to turn ON (1/2/3): 3
Air Conditioner is turned ON  (Cooling started).

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Smart_Home.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Smart_Home.java
SMART HOME DEVICE CONTROL
1. Light
2. Fan
3. Air Conditioner
Choose a device to turn ON (1/2/3): 4
Invalid choice!
```

8. An e-commerce platform calculates discounts differently depending on input: price only, price with membership type, and price with membership plus coupon code. Design a class DiscountCalculator that handles all cases.
Hint: Method overloading is applicable.

JAVA CODE:

Main_Logger.java | Main_String_Valid_Parentheses.java | Main_Payment_System_Process.java | Ma

Main_Discount_Calculator.java > Main_Discount_Calculator

```java
import java.util.Scanner;
class DiscountCalculator {
    double calculateDiscount(double price) {
        return price * 0.05;
    }
    double calculateDiscount(double price, String membershipType) {
        membershipType = membershipType.toLowerCase();
        double rate;
        if (membershipType.equals(anObject: "silver")) rate = 0.10;
        else if (membershipType.equals(anObject: "gold")) rate = 0.15;
        else if (membershipType.equals(anObject: "platinum")) rate = 0.20;
        else rate = 0.05;
        return price * rate;
    }
    double calculateDiscount(double price, String membershipType, String couponCode) {
        double membershipDiscount = calculateDiscount(price, membershipType);
        couponCode = couponCode.toUpperCase();
        double couponDiscount = 0;
        if (couponCode.equals(anObject: "SAVE50")) couponDiscount = 50;
        else if (couponCode.equals(anObject: "OFF10")) couponDiscount = price * 0.10;
        else if (couponCode.equals(anObject: "FREESHIP")) couponDiscount = 0;
        return membershipDiscount + couponDiscount;
    }
}
public class Main_Discount_Calculator {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        DiscountCalculator dc = new DiscountCalculator();
        System.out.println(x: "DISCOUNT CALCULATOR");
        System.out.println(x: "1. Price only");
        System.out.println(x: "2. Price + Membership");
        System.out.println(x: "3. Price + Membership + Coupon");
        System.out.print(s: "Enter choice (1/2/3): ");
        int choice = sc.nextInt();
        sc.nextLine();
        System.out.print(s: "Enter product price: ");
```

Main_Logger.java | Main_String_Valid_Parentheses.java | Main_Payment_System_Process.java | Main_Smart_H

Main_Discount_Calculator.java > Main_Discount_Calculator > main(String[])

```java
public class Main_Discount_Calculator {
    public static void main(String[] args) {
        double price = sc.nextDouble();
        sc.nextLine();
        double discount = 0;
        switch (choice) {
            case 1:
                discount = dc.calculateDiscount(price);
                break;
            case 2:
                System.out.print(s: "Enter membership type (Silver/Gold/Platinum/None): ");
                String membership = sc.nextLine();
                discount = dc.calculateDiscount(price, membership);
                break;
            case 3:
                System.out.print(s: "Enter membership type (Silver/Gold/Platinum/None): ");
                String membership2 = sc.nextLine();
                System.out.print(s: "Enter coupon code (SAVE50/OFF10/FREESHIP/Any): ");
                String coupon = sc.nextLine();
                discount = dc.calculateDiscount(price, membership2, coupon);
                break;
            default:
                System.out.println(x: "Invalid choice!");
                sc.close();
                return;
        }
        double finalPrice = price - discount;
        if (finalPrice < 0) finalPrice = 0;
        System.out.println(x: "\n--- BILL ---");
        System.out.printf(format: "Original Price : ₹%.2f%n", price);
        System.out.printf(format: "Discount       : ₹%.2f%n", discount);
        System.out.printf(format: "Final Price    : ₹%.2f%n", finalPrice);
        sc.close();
    }
}
```

OUTPUT:

```
C:\Windows\System32\cmd.e

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Discount_Calculator.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Discount_Calculator.java
DISCOUNT CALCULATOR
1. Price only
2. Price + Membership
3. Price + Membership + Coupon
Enter choice (1/2/3): 1
Enter product price: 1234

--- BILL ---
Original Price : ?1234.00
Discount       : ?61.70
Final Price    : ?1172.30

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Discount_Calculator.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Discount_Calculator.java
DISCOUNT CALCULATOR
1. Price only
2. Price + Membership
3. Price + Membership + Coupon
Enter choice (1/2/3): 2
Enter product price: 1234
Enter membership type (Silver/Gold/Platinum/None): gold

--- BILL ---
Original Price : ?1234.00
Discount       : ?185.10
Final Price    : ?1048.90

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Discount_Calculator.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Discount_Calculator.java
DISCOUNT CALCULATOR
1. Price only
2. Price + Membership
3. Price + Membership + Coupon
Enter choice (1/2/3): 3
Enter product price: 1234
Enter membership type (Silver/Gold/Platinum/None): gold
Enter coupon code (SAVE50/OFF10/FREESHIP/Any): ANY

--- BILL ---
Original Price : ?1234.00
Discount       : ?185.10
Final Price    : ?1048.90
```

9. A ride-sharing platform supports BikeRide, AutoRide, and CarRide. All inherit from Ride and override calculateFare(distance) with different pricing rules.
Hint: Method overriding is applicable.

JAVA CODE:

Main_String_Valid_Parentheses.java | Main_Payment_System_Process.java | Main_

Main_Ride_Sharing.java > ...

```java
import java.util.Scanner;
class Ride {
    double calculateFare(double distance) {
        return 0;
    }
}
class BikeRide extends Ride {
    @Override
    double calculateFare(double distance) {
        double baseFare = 20;
        double perKm = 8;
        return baseFare + (perKm * distance);
    }
}
class AutoRide extends Ride {
    @Override
    double calculateFare(double distance) {
        double baseFare = 30;
        double perKm = 12;
        return baseFare + (perKm * distance);
    }
}
class CarRide extends Ride {
    @Override
    double calculateFare(double distance) {
        double baseFare = 50;
        double perKm = 18;
        return baseFare + (perKm * distance);
    }
}
public class Main_Ride_Sharing {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "RIDE SHARING FARE CALCULATOR");
        System.out.println(x: "1. Bike Ride");
        System.out.println(x: "2. Auto Ride");
```

Main_String_Valid_Parentheses.java | Main_Payment_System_Process.java | Main_Smart_Hom

Main_Ride_Sharing.java > ...

```java
public class Main_Ride_Sharing {
    public static void main(String[] args) {
        System.out.println(x: "2. Auto Ride");
        System.out.println(x: "3. Car Ride");
        System.out.print(s: "Choose ride type (1/2/3): ");
        int choice = sc.nextInt();
        System.out.print(s: "Enter distance (in km): ");
        double distance = sc.nextDouble();
        Ride ride;
        switch (choice) {
            case 1:
                ride = new BikeRide();
                break;
            case 2:
                ride = new AutoRide();
                break;
            case 3:
                ride = new CarRide();
                break;
            default:
                System.out.println(x: "Invalid choice!");
                sc.close();
                return;
        }
        double fare = ride.calculateFare(distance);
        System.out.println(x: "\n--- FARE DETAILS ---");
        System.out.printf(format: "Distance: %.2f km%n", distance);
        System.out.printf(format: "Total Fare: ₹%.2f%n", fare);
        sc.close();
    }
}
```

OUTPUT:

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Ride_Sharing.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Ride_Sharing.java
error: no class declared in source file

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Ride_Sharing.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Ride_Sharing.java
RIDE SHARING FARE CALCULATOR
1. Bike Ride
2. Auto Ride
3. Car Ride
Choose ride type (1/2/3): 1
Enter distance (in km): 123

--- FARE DETAILS ---
Distance: 123.00 km
Total Fare: ?1004.00

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Ride_Sharing.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Ride_Sharing.java
RIDE SHARING FARE CALCULATOR
1. Bike Ride
2. Auto Ride
3. Car Ride
Choose ride type (1/2/3): 2
Enter distance (in km): 1234

--- FARE DETAILS ---
Distance: 1234.00 km
Total Fare: ?14838.00

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Ride_Sharing.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Ride_Sharing.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Ride_Sharing.java
RIDE SHARING FARE CALCULATOR
1. Bike Ride
2. Auto Ride
3. Car Ride
Choose ride type (1/2/3): 3
Enter distance (in km): 1234

--- FARE DETAILS ---
Distance: 1234.00 km
Total Fare: ?22262.00
```

10. Design a flexible search utility where the same method search() can search an integer in an integer array, a string in a string array, and optionally within a given index range. Hint: Hint: Method overloading is applicable.

JAVA CODE:

a | Main_Payment_System_Process.java | Main_Smart_Home.java | Main_Discount_Ca

Main_Flexible_Search.java > Main_Flexible_Search > main(String[])

```java
import java.util.Scanner;
class SearchUtility {
    int search(int[] arr, int key) {
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] == key) return i;
        }
        return -1;
    }
    int search(int[] arr, int key, int start, int end) {
        if (arr == null || arr.length == 0) return -1;
        if (start < 0) start = 0;
        if (end >= arr.length) end = arr.length - 1;
        if (start > end) return -1;
        for (int i = start; i <= end; i++) {
            if (arr[i] == key) return i;
        }
        return -1;
    }
    int search(String[] arr, String key) {
        if (key == null) return -1;
        for (int i = 0; i < arr.length; i++) {
            if (key.equals(arr[i])) return i;
        }
        return -1;
    }
    int search(String[] arr, String key, int start, int end) {
        if (arr == null || arr.length == 0 || key == null) return -1;
        if (start < 0) start = 0;
        if (end >= arr.length) end = arr.length - 1;
        if (start > end) return -1;
        for (int i = start; i <= end; i++) {
            if (key.equals(arr[i])) return i;
        }
        return -1;
    }
}
```

Main_Payment_System_Process.java | Main_Smart_Home.java | Main_Discount_Calculator.java

se (Ctrl+F4) _Flexible_Search.java > Main_Flexible_Search > main(String[])

```java
public class Main_Flexible_Search {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        SearchUtility su = new SearchUtility();
        System.out.println(x: "FLEXIBLE SEARCH UTILITY");
        System.out.println(x: "1. Search Integer in Integer Array");
        System.out.println(x: "2. Search String in String Array");
        System.out.print(s: "Enter choice (1/2): ");
        int choice = sc.nextInt();
        sc.nextLine();
        System.out.print(s: "Do you want to search within an index range? (yes/no): ");
        String rangeChoice = sc.nextLine().trim().toLowerCase();
        boolean useRange = rangeChoice.equals(anObject: "yes");
        if (choice == 1) {
            System.out.print(s: "Enter array size: ");
            int n = sc.nextInt();
            int[] arr = new int[n];
            System.out.println("Enter " + n + " integers:");
            for (int i = 0; i < n; i++) arr[i] = sc.nextInt();
            System.out.print(s: "Enter integer to search: ");
            int key = sc.nextInt();
            int index;
            if (useRange) {
                System.out.print(s: "Enter start index: ");
                int start = sc.nextInt();
                System.out.print(s: "Enter end index: ");
                int end = sc.nextInt();
                index = su.search(arr, key, start, end);
            } else {
                index = su.search(arr, key);
            }
            if (index != -1) System.out.println("Found at index: " + index);
            else System.out.println(x: "Not found!");
        } else if (choice == 2) {
            System.out.print(s: "Enter array size: ");
            int n = sc.nextInt();
```

a | Main_Payment_System_Process.java | Main_Smart_Home.java | Main_Discount_Calculator.jav

Main_Flexible_Search.java > Main_Flexible_Search > main(String[])

```java
public class Main_Flexible_Search {
    public static void main(String[] args) {
            sc.nextLine();
            String[] arr = new String[n];
            System.out.println("Enter " + n + " strings:");
            for (int i = 0; i < n; i++) arr[i] = sc.nextLine();
            System.out.print(s: "Enter string to search (case-sensitive): ");
            String key = sc.nextLine();
            int index;
            if (useRange) {
                System.out.print(s: "Enter start index: ");
                int start = sc.nextInt();
                System.out.print(s: "Enter end index: ");
                int end = sc.nextInt();
                index = su.search(arr, key, start, end);
            } else {
                index = su.search(arr, key);
            }
            if (index != -1) System.out.println("Found at index: " + index);
            else System.out.println(x: "Not found!");
        } else {
            System.out.println(x: "Invalid choice!");
        }
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Flexible_Search.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Flexible_Search.java
FLEXIBLE SEARCH UTILITY
1. Search Integer in Integer Array
2. Search String in String Array
Enter choice (1/2): 1
Do you want to search within an index range? (yes/no): yes
Enter array size: 10
Enter 10 integers:
1 2 3 4 5 6 7 8 9 10
Enter integer to search: 1
Enter start index: 0
Enter end index: 9
Found at index: 0
```

```
C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>javac Main_Flexible_Search.java

C:\Users\pavan\OneDrive\Pictures\Screenshots\java\ACTIVITY_SHEET_7.java>java Main_Flexible_Search.java
FLEXIBLE SEARCH UTILITY
1. Search Integer in Integer Array
2. Search String in String Array
Enter choice (1/2): 2
Do you want to search within an index range? (yes/no): yes
Enter array size: 5
Enter 5 strings:
p
a
v
a
n
Enter string to search (case-sensitive): p
Enter start index: 0
Enter end index: 4
Found at index: 0
```

5. ASSIGNMENT EVALUATION

0 : Not Done
1 : Incomplete
2 : Late Completion
3 : Needs Improvement
4 : Complete
5 : Well Done

6. VIVA QUESTIONS

1. What is polymorphism in Java?

2. What is method overloading?

3. What is method overriding?

4. What is the difference between overloading and overriding?

5. Which polymorphism occurs at compile time?

6. Which polymorphism occurs at runtime?

7. Can overloaded methods have different return types?

8. Why is method overriding important in object-oriented design?